# IV Bolus Practice 1

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

**a** The data below are plasma concentrations of a new drug at different time points after an intravenous bolus injection of 50 mg to a 70-kg patient. Time (hr) and plasma concentration (mg/L): 0.5 and 2.52; 1.0 and 1.59; 1.5 and 1.00; 2.0 and 0.64; 2.5 and 0.40; 3.0 and 0.25. Using the 1.0 hr and 2.0 hr points, what is the elimination rate constant?

**b** A drug given by intravenous bolus injection has an elimination rate constant of 0.91 $hr^{-1}$. What is its half-life of elimination?

**c** After an intravenous bolus injection of 50 mg, a drug with an elimination rate constant of 0.91 $hr^{-1}$ has a plasma concentration of 0.40 mg/L at 2.5 hours. What is the initial plasma concentration?

**d** A drug given by intravenous bolus injection has an initial plasma concentration of 3.89 mg/L and an elimination rate constant of 0.91 $hr^{-1}$. What is the concentration of drug in the plasma 15 minutes after the dose was given?

**e** A 50 mg intravenous bolus dose produces an initial plasma concentration of 3.89 mg/L in a 70-kg patient. What is the apparent volume of distribution?

**f** A drug has an apparent volume of distribution of 12.85 L and an elimination rate constant of 0.91 $hr^{-1}$ in a patient. What is the total body clearance of this drug in this patient?

**g** A 50 mg intravenous bolus dose of a drug with a volume of distribution of 12.85 L gives a plasma concentration of 0.25 mg/L at 3 hours. What is the amount of drug in the body 3 hours after the drug was administered?

**h** A drug given by intravenous bolus injection has an elimination half-life of 0.76 hours. How much time is required for 99.9% of the drug to be eliminated from the body?